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STATE-OF-THE-ART REVIEW

Selective Use of Nonselective Beta-Blockers

A Tailored Antiarrhythmic Approach

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ABSTRACT

Beta-blockers (BBs) are a cornerstone of antiarrhythmic therapy, yet assuming a uniform "class effect" oversimplifies their diverse pharmacological properties and might compromise efficacy. This review synthesizes evidence comparing the antiarrhythmic effects of different BBs in various clinical settings. While cardioselective agents offer improved tolerability, they often fail to blunt the broad adrenergic surge driving specific arrhythmias. Nonselective BBs (eg, nadolol, propranolol) provide superior efficacy in inherited channelopathies, such as catecholaminergic polymorphic ventricular tachycardia and long QT syndrome, and life-threatening electrical storms. Furthermore, carvedilol is superior to metoprolol for preventing postoperative atrial fibrillation and reducing inappropriate shocks in patients with implantable cardioverter-defibrillators. Conversely, for managing chronic supraventricular tachycardia, cardioselective agents remain preferred. By leveraging distinct BB properties, clinicians can overcome the "one-size-fits-all" approach and tailor therapy to optimize antiarrhythmic efficacy. More research is needed to resolve uncertainties and assist clinicians in adopting a personalized approach to BBs therapy. (JACC Clin Electrophysiol. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

Cardiac arrhythmias represent a major global health burden. Atrial fibrillation (AF), the most common sustained arrhythmia, affects millions worldwide and is associated with a 5-fold increased risk of stroke and a 2-fold increased risk of death.[1-3] Ventricular arrhythmias (VAs), including ventricular tachycardia (VT) and ventricular fibrillation, are a primary cause of sudden cardiac death (SCD),[4] which accounts for one-half of all cardiovascular deaths.[5,6]

The sympathetic nervous system plays a critical role in driving arrhythmogenesis. Excessive catecholaminergic stimulation promotes abnormal automaticity, facilitates re-entrant circuits, and lowers the threshold for life-threatening arrhythmias, particularly in the setting of structural heart disease.[7,8] Consequently, antagonism of beta-adrenergic receptors is a cornerstone of antiarrhythmic therapy.[4]

Beta-blockers (BBs) are Class II agents according to the Vaughan-Williams classification and achieve their primary antiarrhythmic effect by competitively inhibiting catecholamines, thereby blunting the proarrhythmic effects of sympathetic stimulation.[9,10] However, the BB class is not uniform. These drugs

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ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation

BB = beta-blocker

CNS = central nervous system

CPVT = catecholaminergic polymorphic ventricular tachycardia

HCM = hypertrophic cardiomyopathy

HF = heart failure

HFrEF = heart failure with reduced ejection fraction

ICD = implantable cardioverter-defibrillator

IV = intravenous

LQTS = long QT syndrome

MSA = membrane-stabilizing activity

NO = nitric oxide

POAF = postoperative atrial fibrillation

RCT = randomized controlled trial

SCD = sudden cardiac death

SR = sinus rhythm

SVT = supraventricular tachycardia

VA = ventricular arrhythmia

VT = ventricular tachycardia

have a wide range of pharmacologic effects, leading to varying levels of effectiveness depending on the specific clinical situation.[11] This heterogeneity includes differences in cardioselectivity (beta$_1$-selectivity), ancillary properties such as alpha$_1$-receptor blockade, intrinsic sympathomimetic activity, direct effects on cardiac ion channels (membrane-stabilizing activity [MSA]), and distinct pharmacokinetic profiles.

Since the 1990s, clinical practice has shifted toward beta$_1$-selective agents. This transition was driven primarily by improved tolerability, simplified once-daily dosing, and a reduced risk of bronchospasm.[9,11] However, this "selective" evolution may have inadvertently compromised antiarrhythmic efficacy in specific high-risk substrates. By sparing beta$_2$-receptors, selective agents fail to blunt the broad adrenergic surge that characterizes states of extreme sympathetic hyperactivation. Consequently, the "one-size-fits-all" adoption of selective agents entails a trade-off: a gain in tolerability against a potential loss in suppression of critical adrenergic triggers.

The current review analyzed the evidence-based comparisons of the antiarrhythmic effects of different BBs across the spectrum of AF, supraventricular tachycardia (SVT), VA, and specific settings of cardiomyopathies and inherited channelopathies. By synthesizing head-to-head comparative data, we aimed to provide a framework to guide the appropriate choice of BBs according to specific clinical scenarios.

PHARMACOLOGIC HETEROGENEITY OF BETA-ADRENERGIC BLOCKERS

To understand the comparative clinical efficacy of BBs, it is essential first to appreciate their pharmacologic diversity. The antiarrhythmic profile of any given agent is a composite of its primary receptor interactions and its unique ancillary properties. The main differentiating features are summarized in Table 1.

CARDIOSELECTIVITY (BETA$_1$-SELECTIVITY). Second-generation agents such as metoprolol, bisoprolol, and atenolol have a higher affinity for the beta$_1$-receptors concentrated in cardiac tissue than for beta$_2$-receptors found in the bronchioles and peripheral vasculature. In contrast, first-generation agents such as propranolol and nadolol, as well as some third-generation agents such as carvedilol and sotalol, are nonselective, blocking both beta$_1$-receptors and beta$_2$-receptors.[12]

HIGHLIGHTS

- BBs are often used interchangeably despite pharmacologic differences that significantly affect antiarrhythmic efficacy.
- Tailoring BB selection improves management of cardiac arrhythmias in specific clinical scenarios.
- Prospective head-to-head trials are essential to transition from class effect assumptions to precision pharmacology.

VASODILATORY PROPERTIES. Third-generation BBs possess additional vasodilatory properties mediated through distinct mechanisms. Carvedilol and labetalol achieve this through concomitant alpha$_1$-adrenergic blockade, which reduces peripheral vascular resistance.[13] Nebivolol, distinguished by its extremely high beta$_1$-selectivity, induces vasodilation via beta$_3$-adrenergic receptor agonism, stimulating endothelial nitric oxide (NO) synthase and leading to NO-mediated vasodilation.[14]

ANCILLARY ELECTROPHYSIOLOGICAL PROPERTIES. Certain BBs have direct effects on cardiac ion channels, contributing to their antiarrhythmic profile.

- Class III activity: Sotalol combines nonselective beta-blockade with Class III antiarrhythmic activity by blocking the delayed rectifier potassium channels,[15] prolonging the action potential duration and refractory period. The net effect is the prolongation of the QT interval that carries a dose-dependent risk of torsades de pointes.[16]
- MSA: At higher concentrations, propranolol exhibits a "quinidine-like" MSA by blocking voltage-gated sodium channels (Class I effect). This is believed to be a key contributor to its enhanced efficacy against VAs.[17]
- Multi-ion channel blockade: Carvedilol exhibits the broadest electrophysiological profile, inhibiting multiple cardiac ion channels. In addition to the beta-blockade effect (Class II), it blocks the rapid component of the delayed rectifier potassium current, conferring a Class III action,[18] L-type calcium channels (Class IV action), and voltage-gated sodium channels (Class I action).[19]

TABLE 1 Pharmacologic Properties of Beta-Blockers

Agent	Generation	Selectivity	Ancillary Properties	Lipophilicity	Half-Life (h)	Elimination Route	Side Effects/Precautions
Propranolol	First	Nonselective ($beta_1$ = $beta_2$)	MSA	High	3-6	Hepatic (>95%) (<1% renal unchanged)	Bradycardia, hypotension, bronchospasm, masks hypoglycemia, CNS effects (vivid dreams, depression)
Nadolol	First	Nonselective ($beta_1$ = $beta_2$)	None	Low	20-24	Renal (~100%) (not metabolized)	Bradycardia, bronchospasm, fatigue, cold extremities. Requires renal dose adjustment
Sotalol	First	Nonselective ($beta_1$ = $beta_2$)	Class III (potassium channel blockade)	Low	12	Renal (>90%) (not metabolized)	Life-threatening proarrhythmia (torsades de pointes/QT prolongation). Bradycardia, bronchospasm. Strict renal dosing required
Atenolol	Second	$Beta_1$-selective	None	Low	6-7	Renal (>85%) (<10% hepatic)	Bradycardia, fatigue, cold extremities, hypotension. Accumulates in renal failure
Bisoprolol	Second	Highly $beta_1$-selective	None	Moderate	10-12	Renal (50%)/hepatic (50%)	Bradycardia, fatigue, asthenia, dizziness (generally well-tolerated; loss of selectivity at higher doses can trigger bronchospasm)
Metoprolol	Second	$Beta_1$-selective	Weak MSA	Moderate	3-7	Hepatic (>95%) (<5% renal unchanged)	Bradycardia, hypotension, fatigue, dizziness, depression
Esmolol	Second	$Beta_1$-selective	Ultra-short-acting (IV only)	NA	~9 min	RBC esterases (rapid hydrolysis; renal excretion of metabolites' for better readability)	Symptomatic hypotension (most frequent), diaphoresis, injection site reactions (phlebitis, skin necrosis with extravasation)
Carvedilol	Third	Nonselective ($beta_1$ = $beta_2$); $alpha_1$-blockade	Multi-ion channel blockade; antioxidant	High	7-10	Hepatic (>98%) (<2% renal unchanged)	Orthostatic hypotension, dizziness, fatigue, weight gain, bronchospasm. Precaution: intraoperative floppy iris syndrome
Labetalol	Third	Nonselective ($beta_1$ = $beta_2$); $alpha_1$-blockade	Partial $beta_2$ agonism; weak MSA	Moderate	6-8	Hepatic (>95%) (<5% renal unchanged)	Orthostatic hypotension, dizziness, scalp tingling (formication). Warning: Rare but severe hepatocellular injury/hepatotoxicity
Nebivolol	Third	Highest $beta_1$-selectivity	NO-mediated vasodilation	High	10-12	Hepatic (>99%) (active metabolites; <1% renal unchanged)	Headache, fatigue, dizziness, bradycardia
Landiolol	Third	Highest $beta_1$-selectivity	Ultra-short-acting (IV only)	NA	~3-4 min	Plasma esterases (pseudocholinesterase hydrolysis)	Hypotension, bradycardia, injection site reactions (safer hemodynamic profile regarding cardiac depression compared with esmolol)

CNS = central nervous system; IV = intravenous; MSA = membrane-stabilizing activity; NA = not applicable; NO = nitric oxide; RBC = red blood cell.

PHARMACOKINETIC VARIABLES. Lipophilicity and half-life also play a critical role. Lipophilicity affects the central nervous system (CNS) penetration: highly lipophilic agents, such as propranolol, can achieve a central sympatholytic effect but may cause more CNS side effects. Hydrophilic agents such as atenolol and nadolol have limited CNS penetration.[20,21] The long half-life of nadolol (20-24 hours) allows for reliable once-daily dosing, a significant advantage for patient compliance.[22] The ultra-short-acting intravenous (IV) agents esmolol (half-life of approximately 9 minutes) and landiolol (half-life approximately 3-4 minutes) are rapidly cleared, allowing for precise titration and rapid reversal of effects in acute care settings.[23,24]

Finally, the route of elimination (hepatic metabolism vs renal excretion) is a determinant of safety in patients with comorbidities. Hydrophilic agents such as sotalol, nadolol, and atenolol are primarily eliminated unchanged by the kidneys. Consequently, their dosage must be adjusted in patients with renal insufficiency to avoid drug accumulation; this is particularly critical for sotalol, in which reduced clearance increases the risk of dose-dependent QT prolongation and torsades de pointes. Conversely, lipophilic agents such as carvedilol, metoprolol, and propranolol undergo extensive hepatic metabolism. Although this makes them safer in cases of renal failure, their plasma levels can be influenced by liver

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function and genetic polymorphisms in cytochrome P450 enzymes (eg, CYP2D6 for metoprolol), leading to variable therapeutic responses. Bisoprolol occupies a unique middle ground with a "balanced" clearance profile (approximately 50% renal and 50% hepatic), offering a predictable safety profile in patients with mild to moderate dysfunction of either organ.

COMPARATIVE EFFICACY IN AF

ACUTE VENTRICULAR RATE CONTROL. In the acute setting, landiolol offers the most favorable combination of efficacy in heart rate control, preservation of blood pressure, and short half-life. Indeed, a meta-analysis confirmed that landiolol provides superior heart rate control with a less negative impact on blood pressure and cardiac output compared with nonselective and other beta$_1$-selective BBs, making it a safer option in compromised patients.[25] Also, compared with other short-acting BBs such as esmolol, landiolol has a shorter half-life (3-4 minutes) and significantly higher beta$_1$-selectivity.[23,24]

PREVENTION OF POSTOPERATIVE AF. In the setting of postoperative AF (POAF) prevention, randomized controlled trials (RCTs) and meta-analyses have highlighted important differences among BBs. Among them, carvedilol has consistently shown superior efficacy compared with metoprolol.[26-28] In keeping with this, one meta-analysis reported that patients taking metoprolol had a 59% higher risk of developing POAF compared with those taking carvedilol (risk ratio: 1.59; 95% CI: 1.20-2.12).[27] This superiority is attributed to carvedilol's multiaction profile, including its anti-inflammatory and antioxidant properties, which directly counteract the underlying pathophysiology of POAF.[29,30]

In addition to these 2 agents, other BBs have been investigated in this context. A pilot study found that a perioperative landiolol regimen followed by oral bisoprolol was effective in preventing POAF compared with the control group.[31] Similarly, a prospective RCT comparing nebivolol with metoprolol showed equivalent efficacy between the 2, suggesting that pharmacodynamic nuances may influence results depending on patient characteristics and surgical context.[32]

Overall, these findings support the view that not all BBs exert the same protective effect against POAF and that agents with pleiotropic actions, such as carvedilol, may provide additional benefit in the perioperative management of cardiac surgery patients.

AF IN HEART FAILURE. In patients with heart failure (HF), the relationship between different BBs and atrial arrhythmias has been extensively studied, revealing distinct clinical implications depending on the agent used. In a large pooled analysis of >4,000 patients with HF and implantable cardioverter-defibrillators (ICDs), carvedilol was associated with a 35% lower risk of atrial tachyarrhythmias and inappropriate ICD shocks (often triggered by AF) compared with metoprolol.[33] This observation aligns with the evidence supporting carvedilol's broader pharmacologic profile, which may offer protection through nonselective beta-blockade and mechanisms extending beyond beta$_1$-blockade alone. In contrast, comparative evidence between bisoprolol and carvedilol suggests a differential clinical effect. A prospective RCT found that bisoprolol was more effective than carvedilol in preventing new-onset AF in the subacute period after hospital discharge following coronary artery bypass grafting in patients with HF.[34] This suggests that a strong beta$_1$-blockade may be more crucial than pleiotropic effects once acute inflammation subsides.

Furthermore, a retrospective study of patients with severe HF and AF showed that a significantly higher proportion of patients treated with bisoprolol converted to sinus rhythm (SR) compared with those taking carvedilol (48% vs 16%).[35] However, a landmark meta-analysis of individual-patient data concluded that although BBs reduce mortality in patients with HF with reduced ejection fraction (HFrEF) and SR, they do not provide a clear survival benefit in patients with concomitant AF, underscoring the complexity of managing this dual pathology.[36]

MAINTENANCE OF SR. An RCT comparing sotalol vs bisoprolol for maintaining SR after AF electrical cardioversion found equal efficacy in preventing AF recurrence (41% and 42%, respectively).[37] However, the safety profiles were markedly different: life-threatening proarrhythmia (torsades de pointes) occurred in 3.1% of patients taking sotalol, with no such events in the bisoprolol group. These data show that bisoprolol can be a safer alternative to sotalol with equivalent efficacy. Another RCT compared carvedilol and bisoprolol in 90 patients with persistent AF after cardioversion. Results showed similar efficacy despite a trend favoring carvedilol: 46% of patients taking bisoprolol and 32% of patients taking carvedilol relapsed over a year ($P = 0.486$).[38] Interestingly, evidence from the LIFE (Losartan Intervention For End Point Reduction in Hypertension) trial found that an atenolol-based regimen was

TABLE 2 Comparative Efficacy of Beta-Blockers in AF

Clinical Setting	Study Design	Population	Comparison	Superior Agent	Key Finding
Acute rate control[25]	Meta-analysis of RCTs	N = 1,152 participants (12 RCTs) with acute AF/ flutter	Landiolol vs esmolol and metoprolol	Landiolol	Higher beta$_1$-selectivity and shorter half-life provide superior heart rate control with less hemodynamic compromise
Prevention of postoperative AF[26-28]	3 Meta-analyses of RCTs	N = 765 (Wang et al[26]); N = 1,570 (Norhayati et al[27]); N = 4,452 (Burgess et al[28]) Patients undergoing major cardiac surgery	Carvedilol vs metoprolol	Carvedilol	Significantly lower incidence of POAF attributed to pleiotropic anti-inflammatory and antioxidant effects
Prevention of postdischarge AF (post-CABG in HF)[34]	RCT	N = 320. Patients with HFrEF recently discharged post-CABG	Bisoprolol vs carvedilol	Bisoprolol	More effective at preventing new-onset AF in the subacute phase (14.6% vs 23%)
SR conversion in severe HF[35]	Retrospective Cohort	N = 83. Patients with HFrEF and AF	Bisoprolol vs carvedilol	Bisoprolol	Associated with a significantly higher rate of conversion from AF to SR (48% vs 16%)
AF with HF (ICD patients)[33]	Pooled Analysis/ observational	N = 4,194. Patients with HFrEF and an implanted ICD	Carvedilol vs metoprolol	Carvedilol	Associated with a 35% lower risk of atrial tachyarrhythmias and inappropriate ICD shocks (HR: 0.65)
Maintenance of SR (post-cardioversion)[37]	RCT	N = 128. Patients with persistent AF undergoing successful electrical cardioversion	Bisoprolol vs sotalol	Bisoprolol (safer)	Equal efficacy in preventing AF recurrence but with a significantly better safety profile (3.1% torsades de pointes with sotalol)
Maintenance of SR (post-cardioversion)[38]	RCT	N = 90. Patients with persistent AF post-cardioversion	Bisoprolol vs carvedilol	Carvedilol = bisoprolol	Equal efficacy in preventing AF recurrence, with a (nonsignificant) trend favoring carvedilol
Prevention of new-onset AF (hypertension)[39]	RCT	N = 9,193 (LIFE trial). High-risk hypertensive patients	Atenolol vs losartan	Losartan	Atenolol-based regimen was associated with a higher risk of developing new-onset AF compared with losartan

AF = atrial fibrillation; CABG = coronary artery bypass graft; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; ICD = implantable cardioverter-defibrillator; POAF = postoperative atrial fibrillation; RCT = randomized controlled trial; SR = sinus rhythm.

associated with a significantly higher risk of developing new-onset AF compared with a losartan-based regimen despite similar blood pressure control.[39] Although the exact mechanism is unclear, it can be speculated that its hydrophilic nature and lack of central sympatholytic effect could result in relatively unopposed peripheral beta$_2$-stimulation or an inadequate blunting of central sympathetic drive, particularly in a high-risk hypertensive population. Conversely, losartan advantages could be linked to the atrial antiremodeling effects of angiotensin receptor blockers (Table 2).

COMPARATIVE EFFICACY IN SVTs

ACUTE MANAGEMENT OF SVT. IV BBs are a second-line therapy recommended when vagal maneuvers and adenosine fail.[40] In critically ill or hemodynamically unstable patients, ultra-short-acting agents are preferred. An RCT comparing IV esmolol vs IV propranolol found comparable efficacy in terminating SVTs (72% vs 69%), but the effects of esmolol dissipated within 10 minutes of cessation, highlighting its superior safety and titrability.[41] Landiolol is a premier choice in fragile patients due to its shorter half-life and greater beta$_1$-selectivity, providing superior heart rate control with greater hemodynamic stability compared with esmolol.[42] In hemodynamically stable patients, IV metoprolol is a guideline-preferred agent for the acute treatment of SVT. IV propranolol is also an effective alternative.[40,43]

PREVENTION OF SVT RELAPSES. For long-term prevention of SVT recurrences, oral BBs are a Class I recommendation for patients who are not candidates for or prefer not to undergo catheter ablation.[40,43] Comparative efficacy data are limited, and the choice is often guided by pharmacokinetic variables and side-effect profiles. Cardioselective agents (eg, metoprolol, bisoprolol) are usually preferred first-line drugs due to their targeted cardiac effects and favorable tolerability.[21] Nadolol's very long half-life (20-24 hours) allows for reliable once-daily dosing,[22] making it particularly useful in pediatric populations in which compliance is challenging.[44] Propranolol is a well-established option but often

TABLE 3 Comparative Efficacy of Beta-Blockers in SVTs

Clinical Setting	Study Design	Population	Comparison	Superior Agent	Key Finding
Acute rate control[42]	PSM study	N = 438. Critically ill patients	Landiolol vs esmolol	Landiolol	Shorter half-life, higher $beta_1$-selectivity, and less negative impact on blood pressure and contractility
Acute termination (stable patient)[41]	RCT	N = 127. Hemodynamically stable patients presenting with spontaneous acute SVT	Esmolol vs propranolol	Esmolol (safer)	Comparable efficacy (72% vs 69%), but esmolol has a much faster offset, allowing for rapid reversal of adverse effects
Acute termination (stable patient)[40]	Clinical guidelines	NA (clinical guidelines)	Metoprolol (IV)	Guideline preferred	Effective and widely used standard agent for hemodynamically stable SVT
Chronic prevention (pediatrics)[44]	Retrospective cohort	N = 28. Infants and pediatric patients with recurrent SVT in whom daily drug adherence is challenging	Nadolol	Preferred Agent	Very long half-life allows for convenient and reliable once-daily dosing, improving compliance
Chronic prevention (adults)[40]	Clinical guidelines	NA (clinical guidelines)	Cardioselective agents (metoprolol, bisoprolol, atenolol)	Preferred first-line	Favorable tolerability and targeted cardiac effects. Atenolol may cause fewer CNS side effects

PSM = propensity score matching; SVT = supraventricular tachycardia; other abbreviations as in Tables 1 and 2.

requires multiple daily doses and can cause more CNS side effects due to its high lipophilicity (Table 3).[21]

COMPARATIVE EFFICACY IN VAs, CARDIOMYOPATHIES, AND CHANNELOPATHIES

ACUTE MANAGEMENT OF VAs. In the acute management of life-threatening VAs, particularly electrical storm, BB selection has a key impact on outcomes. A landmark double-blind RCT found that oral propranolol was significantly superior to oral metoprolol in patients with electrical storm already receiving amiodarone.[45] Indeed, propranolol resulted in a markedly lower incidence of recurrent VAs and shorter hospital stays. This superiority is attributed to propranolol's unique combination of nonselective $beta_1$/$beta_2$-blockade, high lipophilicity that allows a central sympatholytic effect, and MSA properties that metoprolol lacks.[46]

Complementary evidence comes from case reports and small series describing patients with severe, recurrent VT unresponsive to the $beta_1$-selective bisoprolol who achieved complete arrhythmia suppression after switching to propranolol.[47] These observations emphasize the relevance of broad-spectrum adrenergic blockade and central modulation of sympathetic drive as key determinants of efficacy in the management of refractory VAs.

VAs IN STRUCTURAL HEART DISEASE. LV Dysfunction of Ischemic and Nonischemic Origin. In patients with HFrEF, the evidence supports carvedilol, metoprolol succinate, and bisoprolol for mortality reduction.[48-50] Indeed, a large network meta-analysis found no difference in all-cause mortality between carvedilol and metoprolol, suggesting that the mortality benefit in HFrEF is largely a class effect of beta-blockade.[51] However, when considering arrhythmic endpoints, comparative data among agents are less uniform. Early RCTs of metoprolol vs sotalol yielded conflicting results. Pacifico et al[52] reported a lower incidence of ICD shocks with sotalol compared with conventional therapy (which included BBs in 73% of patients). In contrast, Seidl et al[53] reported the opposite, showing superior prevention of fast VT or ventricular fibrillation recurrence with metoprolol in patients with ICDs (80% vs 46% event-free survival at 2 years). Kettering et al[54] subsequently found no significant difference between the 2 agents, suggesting that their relative benefits may depend on patient selection and arrhythmia substrate.

Among BBs with proven survival benefit, COMET (Carvedilol Or Metoprolol European Trial) showed a significant advantage of carvedilol over metoprolol tartrate in all-cause and cardiovascular mortality.[55,56] When focusing on arrhythmia endpoints, a large pooled analysis of >4,000 ICD patients found that carvedilol was associated with a 35% lower risk of atrial tachyarrhythmias and inappropriate shocks compared with metoprolol, along with a nonsignificant trend toward fast VA reduction (HR: 0.84; 95% CI: 0.70-1.02).[33] In line with this, a smaller retrospective analysis found that carvedilol was associated with a lower risk of appropriate ICD shocks than metoprolol succinate,[57] and a small RCT showed that in patients with dilated cardiomyopathy who were poor responders to metoprolol, switching to carvedilol improved left ventricular ejection fraction and reduced ventricular ectopy.[58] A meta-analysis of RCTs directly comparing carvedilol vs beta$_1$-selective agents further confirmed that carvedilol was associated with a significant reduction in sustained VAs and appropriate ICD therapies.[59]

Finally, evidence supports the notion that dose optimization plays a critical role in arrhythmia prevention. Higher BB dosages are associated with a dose-dependent increase in the effectiveness of antitachycardia pacing, thereby reducing ICD shocks.[60] This principle remains a key factor for the successful management of VAs, regardless of the specific agent used.

Chronic Coronary Syndromes. In the modern reperfusion era, the routine long-term use of BBs in patients with chronic coronary syndrome (stable coronary artery disease) and preserved ejection fraction is increasingly debated. Large registry data indicate that beyond the first year post-myocardial infarction, BBs may not confer a survival benefit in this population.[61,62] Regarding arrhythmic risk, evidence specifically showing a reduction in SCD in chronic coronary syndrome patients with preserved ejection fraction is lacking. In the absence of scar-related VT or significant ventricular ectopy, the antiarrhythmic yield of BBs here is low. Therefore, the choice of agent is driven by antianginal efficacy and tolerability rather than arrhythmia suppression. In this context, beta$_1$-selective agents or vasodilating agents (nebivolol) are preferred to maximize exercise tolerance and minimize fatigue.

Takotsubo Syndrome. Given that a massive catecholamine surge is central to the pathophysiology of Takotsubo syndrome, BB therapy is naturally an appealing treatment strategy. However, the evidence is nuanced and requires a phase-dependent approach. In the acute phase, the choice of agent is critical if dynamic left ventricular outflow tract obstruction is present. In this scenario, vasodilating BBs (eg, carvedilol, nebivolol) may be deleterious because a reduction in afterload can worsen the left ventricular outflow tract obstruction.[63] Therefore, in cases of dynamic left ventricular outflow tract obstruction, non-vasodilating agents with a short half-life (eg, esmolol and landiolol) are preferred to reduce contractility without exacerbating the obstruction.[64]

Regarding long-term management, registry data have shown that BB can be used with potential survival benefit, although chronic BB therapy does not reliably prevent recurrence of Takotsubo syndrome.[65,66] Consequently, the routine long-term use of BBs solely for recurrence prevention is not supported by current evidence, and their prescription should be guided by other indications (eg, hypertension, residual HFrEF).

Cardiomyopathies. In cardiomyopathies, BBs remain a cornerstone of management, although their selection and role vary depending on the underlying substrate. In hypertrophic cardiomyopathy (HCM), BBs are the first-line therapy for symptomatic patients. Commonly used agents include metoprolol, propranolol, and atenolol, although robust comparative trials are lacking. Critically, vasodilating BBs (eg, carvedilol, labetalol) should be avoided in patients with the obstructive form of HCM, as reductions in peripheral resistance can worsen outflow tract obstruction.[67]

In arrhythmogenic right ventricular cardiomyopathy, BBs are also recommended as first-line therapy to reduce the risk of VA and SCD,[4] with non-vasodilating agents preferred to reduce the reduction in venous return to a dysfunctional right ventricle[68]; data specifically validating this hemodynamic hypothesis in arrhythmogenic right ventricular cardiomyopathy are limited, however. Sotalol, owing to its additional Class III antiarrhythmic properties, is frequently adopted as a second-line agent.[4,69]

Patients with *LMNA* mutation-related cardiomyopathy represent a unique high-risk subgroup characterized by a dual propensity for conduction system disease and life-threatening ventricular tachyarrhythmias.[70] In this setting, BBs are standard therapy, but the high risk of atrioventricular block complicates the choice. Unlike in classical HFrEF, in which maximizing the dose is the goal, in *LMNA* carriers, BB titration requires vigilant monitoring of the PR interval and bradycardia. Although no single agent has proven superior, the high burden of AF (often the first manifestation) and VT suggests that nonselective agents might offer theoretical advantages for rhythm control. However, their safety regarding conduction slowing must be carefully weighed against the risk of bradyarrhythmias before pacemaker/ICD implantation.[4]

CHANNELOPATHIES. In channelopathies, the efficacy and safety of BBs vary markedly across syndromes. In catecholaminergic polymorphic VT (CPVT), nonselective BBs are unequivocally superior to beta$_1$-selective agents in reducing VA and SCD risk in CPVT. A large international cohort study found

TABLE 4 Comparative Efficacy of BBs in Ventricular Arrhythmias and Sudden Cardiac Death

Clinical Setting	Study Design	Population	Comparison	Superior Agent	Key Finding
Electrical storm[45]	RCT	N = 60. ICD patients experiencing electrical storm, already refractory to amiodarone	Propranolol vs metoprolol	Propranolol	Significantly superior in reducing recurrent VA; attributed to nonselective blockade, high lipophilicity (central effect), and MSA
Secondary prevention (HFrEF/ICD-mortality)[51]	Network Meta-analysis of RCTs	N = 23,122 (21 RCTs included). Patients with HFrEF	Carvedilol vs metoprolol vs bisoprolol	Class effect	Network meta-analysis found no single agent superior to others for reducing all-cause mortality
Secondary prevention (HFrEF/ICD-AF/ inappropiate shocks/VA)[33]	Pooled analysis/ observational	N = 4,194. Patients with HFrEF and an ICD	Carvedilol vs metoprolol	Carvedilol	35% atrial tachycardias and inappropriate ICD shock reduction with carvedilol. Nonsignificant trend favoring carvedilol also for fast VA reduction
Secondary prevention (HFrEF/ICD-any shock)[94]	RCT	N = 412. ICD patients with a recent episode of spontaneous VT/VF	Amiodarone + BB vs sotalol vs BB alone	Amiodarone + BB	Amiodarone + BB was superior to sotalol, which was only marginally better than BB alone
HCM obstructive[67]	Clinical guidelines	NA (clinical guidelines)	Non-vasodilating vs vasodilating BBs	Non-vasodilating	Vasodilating agents (eg, carvedilol, labetalol) should be avoided in obstructive HCM as they can worsen the outflow tract gradient
CPVT[71]	Multicenter cohort study	N = 218. Symptomatic children and young adults diagnosed with CPVT	Nonselective vs selective agents	Nonselective (nadolol/ propranolol)	$Beta_1$-selective blockers associated with a 2-fold higher risk of arrhythmic events compared with nonselective agents
LQTS[74,75,78]	Meta-analysis, network meta-analysis, and systematic review of observational studies	Ahn et al[74] (2017): N = 9,727 Han et al[75] (2020): N = 11,641 Went et al[78] (2021): Systematic review of 11 publications; patients with congenital LQT1, LQT2, and LQT3 genotypes	Nonselective vs selective agents	Nadolol (in all genotypes)	Nadolol is the most effective agent across LQT1, LQT2, LQT3, and genotype elusive. In LQT3, it significantly reduces cardiac events, specifically by 83% in female subjects. Propranolol shows significant risk reduction in LQT1 but is ranked lower than nadolol in overall efficacy, whereas metoprolol is consistently identified as the least effective agent

BB = beta-blocker; CPVT = catecholaminergic polymorphic ventricular tachycardia; HCM = hypertrophic cardiomyopathy; LQTS = long QT syndrome; VT/VF = ventricular tachycardia/ventricular fibrillation; other abbreviations as in Tables 1 and 2.

that selective $beta_1$-blockers (atenolol, metoprolol, and bisoprolol) were associated with a 2-fold higher risk of arrhythmic events compared with nonselective BBs.[71] Nadolol and propranolol are the agents of choice. A crossover study further showed nadolol to be significantly more effective than selective agents in blunting maximal heart rate during exercise, thereby narrowing the "arrhythmic window."[72]

In long QT syndrome (LQTS), BBs are first-line therapy for reducing VA and SCD risk. Although the management of genotype-negative LQTS remains challenging due to variable phenotypes,[73] guidelines and expert consensus recommend nadolol and propranolol as the preferred agents.[4] A meta-analysis by Ahn et al[74] found that nadolol was effective in reducing cardiac events in both LQT1 and LQT2. At the same time, propranolol and atenolol showed benefit mainly in LQT1, and metoprolol showed no significant benefit in either genotype. These results were corroborated and expanded by a network meta-analysis that ranked nadolol as the most effective agent for preventing cardiac events in both LQT1 and LQT2 genotypes.[75] Of note, an observational study found a higher rate of breakthrough arrhythmic events with atenolol than with other BBs.[76] Although LQT3 was previously thought to be less responsive to adrenergic blockade, contemporary evidence confirms that BBs (especially nadolol) significantly reduce cardiac events also in this population.[77,78]

In contrast to other channelopathies, BBs should generally not be initiated in patients with Brugada syndrome. Propranolol, due to its sodium channel-blocking effect, should be preferably avoided,[79] whereas other BBs might increase vagal tone and slow heart rate, generating a physiological state that favors VA trigger in patients with Brugada syndrome (Table 4).[4,80]

SPECIAL POPULATIONS

PREGNANCY AND LACTATION. Managing arrhythmias during pregnancy requires a delicate balance between maternal safety and fetal well-being. BBs cross the placenta and have been associated with fetal growth restriction, neonatal hypoglycemia, and bradycardia.[81-84] Although beta$_1$-selective agents are often preferred for maternal hypertension due to their safety profile, inherited channelopathies present a unique challenge. Recent evidence suggests that for pregnant women with LQTS, nonselective agents such as nadolol or propranolol remain the most effective therapy for preventing maternal VAs, despite the fetal risk of growth restriction and neonatal adverse effects such as hypoglycemia and bradycardia risks.[85,86]

PEDIATRIC POPULATION. In children, compliance is the primary barrier to effective long-term arrhythmia management. Although propranolol is highly effective, its short half-life necessitates multiple daily doses (3-4 times daily), leading to poor adherence. Nadolol, with its long half-life enabling once-daily dosing, is increasingly favored for pediatric CPVT and LQTS to ensure consistent therapeutic coverage.[87]

ELDERLY PATIENTS. Age-related reduced BB sensitivity and conduction system disease make dose titration critical.[88] The choice of agent must independently consider receptor selectivity and lipophilicity. To minimize CNS side effects (eg, cognitive dysfunction, delirium), hydrophilic agents (eg, atenolol, nadolol) are preferred over lipophilic ones (eg, propranolol) that readily cross the blood-brain barrier.[89] Simultaneously, beta$_1$-selective blockers are favored to avoid bronchospasm in patients with chronic obstructive pulmonary disease. Notably, these properties are not linked: nadolol is nonselective but hydrophilic, which reduces the risk of CNS toxicity, whereas a selective agent such as nebivolol is lipophilic, but its unique NO-mediated effects may help treat isolated systolic hypertension in this population.[90]

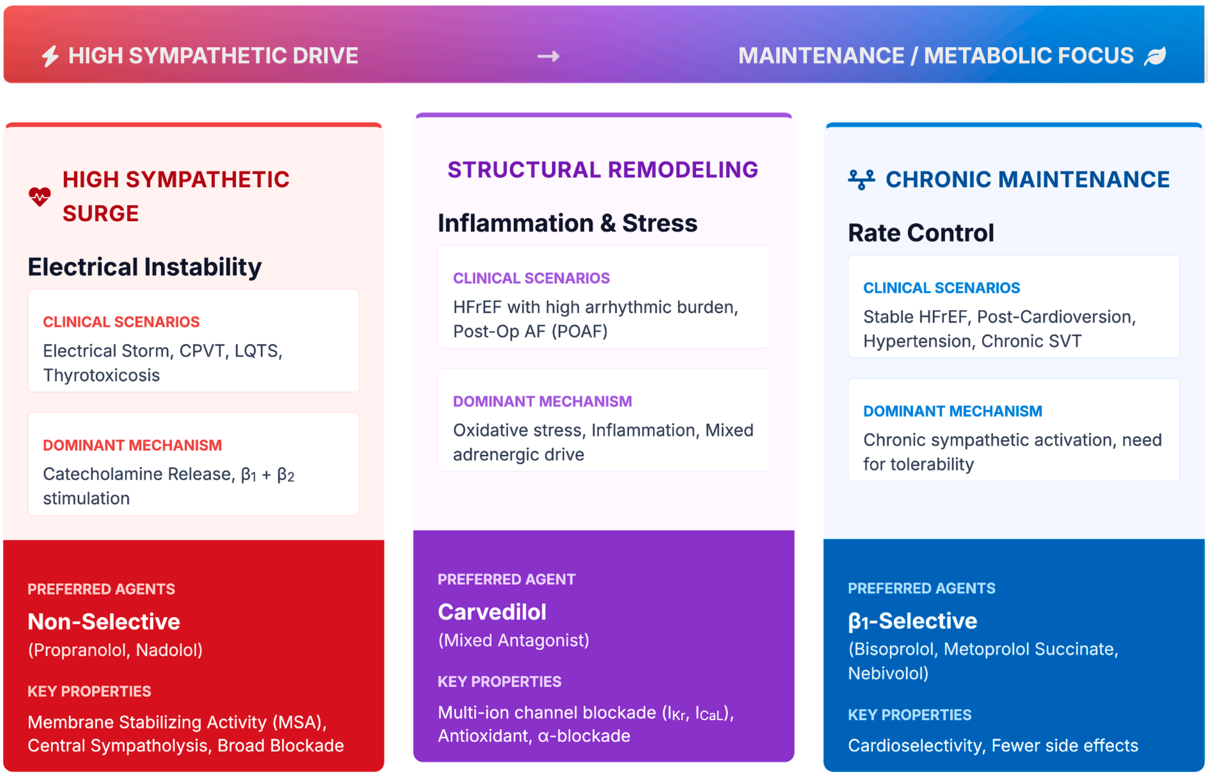
DISCUSSION

A pursuit of cardioselectivity has driven the evolution of BB therapy over the last 3 decades. This shift was motivated by the desire to minimize extra-cardiac side effects. Although this “selective” evolution succeeded in making BBs one of the most prescribed drug classes globally, it inadvertently created a “blind spot” in arrhythmia management. By prioritizing tolerability, clinical practice drifted away from the broad-spectrum adrenergic blockade necessary to treat specific arrhythmia scenarios. Current evidence suggests that, for rate control and hypertension, selectivity is a virtue; however, for suppressing the “adrenergic storm” of CPVT, electrical storm, or complex VAs, the specific ancillary properties of nonselective agents (MSA, central sympatholysis, and potassium-channel blockade) might add value. A thorough analysis of the comparative antiarrhythmic effectiveness of BBs highlights the importance of avoiding BB prescription inertia while emphasizing the need to select the best agent whenever possible (**Figure 1**).[91]

The most compelling finding is the consistent superiority of agents with broader mechanisms of action in high-stress, adrenergically driven arrhythmias. This is evident in inherited channelopathies (eg, CPVT, LQTS), in which nadolol and propranolol are clearly superior beta$_1$-selective blockers.[71,74,92] The superiority of propranolol over metoprolol for electrical storm further reinforces this principle in states of extreme sympathetic surge.[45,47] Of note, propranolol’s ancillary properties extend beyond comprehensive beta$_1$/beta$_2$-blockade (ie, high lipophilicity/central effect) and MSA due to voltage-gated sodium channel blockade, a Class I effect.[46] This mechanism contrasts with carvedilol’s multi-ion channel blockade (Class I, III, and IV activity), which may be critical in its superiority for preventing POAF. Indeed, carvedilol’s combined anti-inflammatory, antioxidant, and broad antiadrenergic properties seem to provide a more comprehensive therapeutic strategy against the inflammatory and oxidative stress that drives POAF,[27,28] whereas the highly selective bisoprolol seems superior for AF prevention in the subacute postdischarge period, when inflammation subsides.[34]

For VA suppression in patients with structural heart disease and ICDs, the evidence remains limited and complex. Although the critical endpoint of all-cause mortality in HFrEF is essentially a class effect,[51] essential distinctions emerge when focusing on arrhythmia-specific outcomes. The superior mortality benefit of carvedilol in COMET (Carvedilol Or Metoprolol European Trial)[55] must be interpreted cautiously due to the comparison with the short-acting metoprolol tartrate, which is not the modern standard (ie, metoprolol succinate).[56] More recent data suggest that carvedilol’s primary advantage in the ICD population is its 35% reduction in atrial tachyarrhythmias and inappropriate shocks, with a nonsignificant trend for reducing fast VAs.[33,93] Furthermore, the OPTIC (Optimal Pharmacological Therapy in Cardioverter Defibrillator Patients) trial showed that, for preventing ICD shocks of any kind, the combination of amiodarone plus a BB was superior to sotalol, which, in turn, displayed only a nonsignificant trend toward superiority over BB therapy alone.[94]

FIGURE 1 **Adrenergic Spectrum of BB Efficacy**



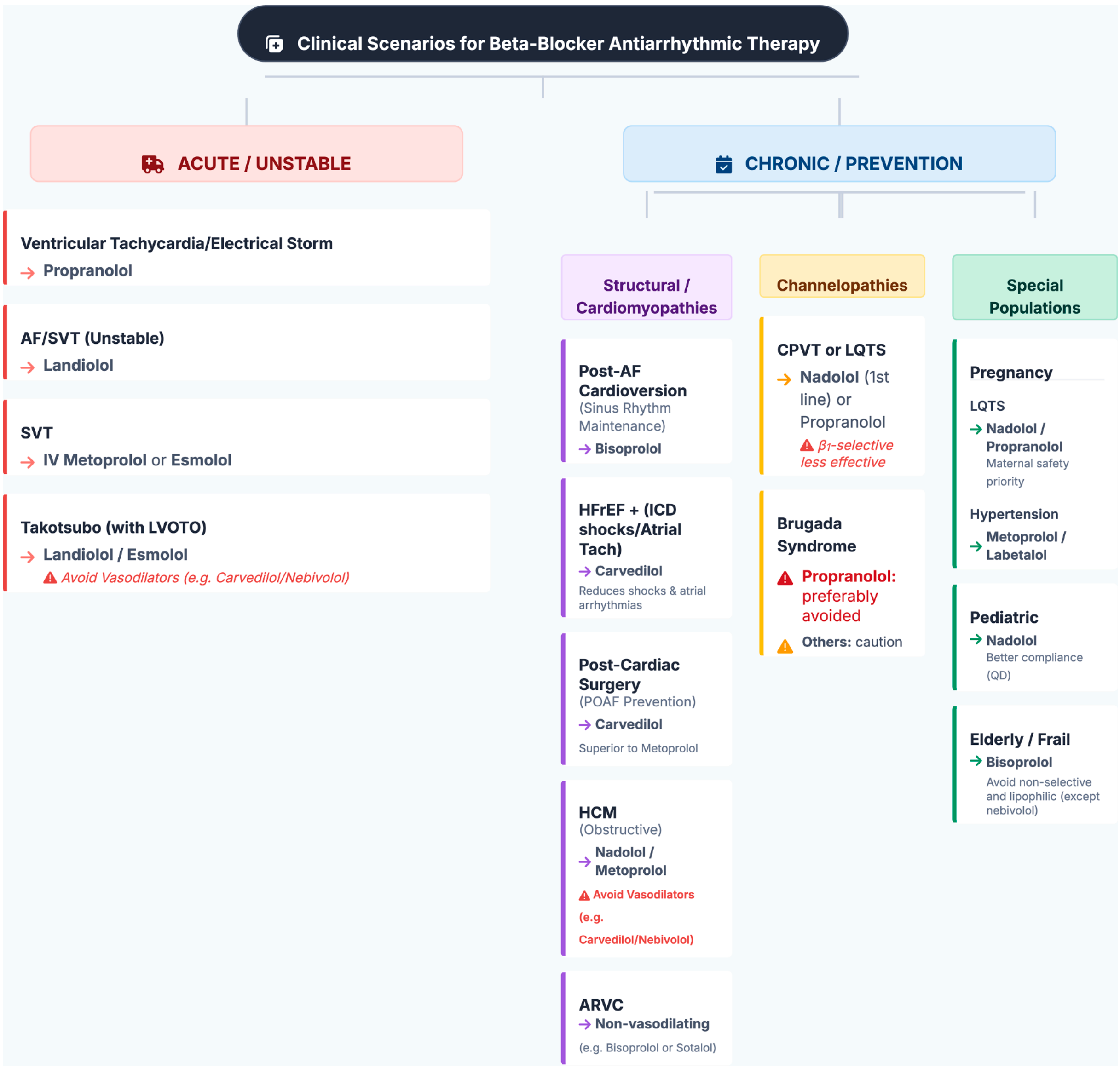
The choice of beta-blocker (BB) should align with the underlying pathophysiological substrate. Conditions characterized by extreme sympathetic surges (eg, electrical storm, catecholaminergic polymorphic ventricular tachycardia [CPVT]) require nonselective agents (eg, propranolol, nadolol) to blunt both beta$_1$- and beta$_2$-mediated triggers. Conditions driven by inflammatory and oxidative stress, such as postoperative atrial fibrillation (AF), benefit from the pleiotropic effects of carvedilol. Conversely, for chronic rhythm maintenance and rate control, in which tolerability is paramount, beta$_1$-selective agents are preferred. HFrEF = heart failure with reduced ejection fraction; LQTS = long QT syndrome; SVT = supraventricular tachycardia.

The conflicting results for VA suppression in HFrEF likely reflect heterogeneity across studies, including observational designs susceptible to confounding by indication and clinical trials using different drug formulations (eg, metoprolol tartrate vs succinate). An often-overlooked clinical pearl is the importance of dose optimization; higher indexed doses of any BB have been shown to increase the efficacy of antitachycardia pacing, thereby reducing shocks,[60] a factor that transcends agent-specific comparisons. The finding that, in the LIFE trial, atenolol was less efficacious than losartan in preventing new-onset AF despite a similar blood pressure control is noteworthy. It may be related to its highly hydrophilic nature and lack of central sympatholytic effect, potentially highlighting its limitations compared with other BBs or associated with the atrial antiremodeling properties of angiotensin receptor blockers.[39]

Of note, in daily clinical practice, clinicians frequently encounter phenotypic “overlap” scenarios in which evidence is scarce. In these cases, BB therapy should be pragmatically tailored based on the patient’s predominant phenotype. For instance, the coexistence of LQTS and HFrEF presents a balance between the arrhythmic phenotype (favoring nadolol or propranolol) and the hemodynamic phenotype (favoring HF-approved agents such as bisoprolol, metoprolol, or carvedilol). In this specific scenario, carvedilol might theoretically represent an appropriate therapeutic bridge; despite the absence of prospective data proving a survival benefit in LQTS, it might provide the nonselective beta$_1$/beta$_2$-blockade required for the LQTS substrate while maintaining the proven mortality benefits essential for HFrEF management.

Finally, the drug safety profile of agents should also guide therapeutic selection. For maintaining SR

FIGURE 2 Algorithm for Tailored BB Selection

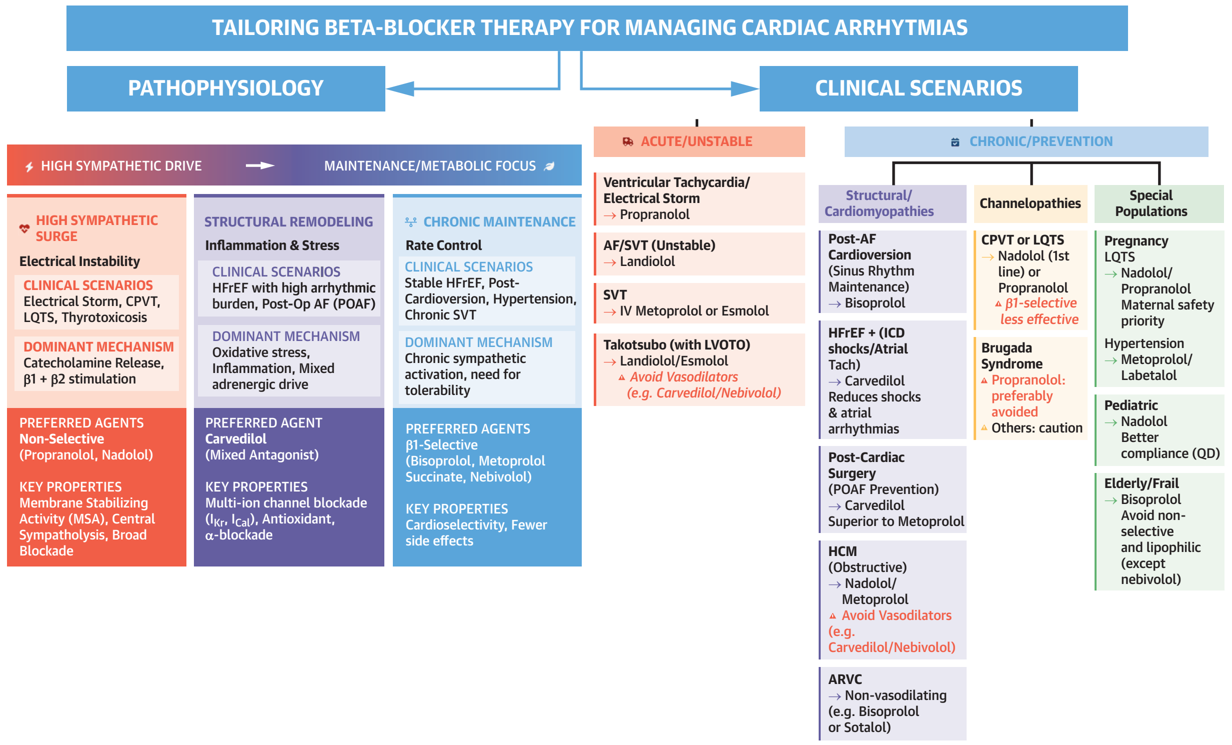


A decision tree guiding the selection of BBs based on specific clinical scenarios, underlying cardiomyopathy, and patient demographic characteristics. The algorithm distinguishes between acute and chronic management and highlights contraindicated agents for specific substrates (eg, vasodilators in obstructive hypertrophic cardiomyopathy [HCM], BBs in Brugada syndrome). ARVC = arrhythmogenic right ventricular cardiomyopathy; ICD = implantable cardioverter-defibrillator; IV = intravenous; LVOTO = left ventricular outflow tract obstruction; POAF = postoperative atrial fibrillation; QD = once daily; Tach = tachycardia; other abbreviations as in Figure 1.

after cardioversion, bisoprolol and carvedilol showed a comparable efficacy without evidence of proarrhythmia, further supporting their use in long-term rhythm control. In contrast, sotalol was associated with a 3.1% incidence of torsades de pointes, a risk not observed with bisoprolol.[37] Similarly, in the acute management of AF with rapid ventricular response, a recent meta-analysis found that landiolol provided superior heart rate control with a better safety profile, causing less hypotension and bradycardia than nonselective BBs.[25]

Finally, there is a paucity of clinical trial data on the antiarrhythmic properties of newer agents, such as nebivolol; a subanalysis of the SENIORS (Study of

CENTRAL ILLUSTRATION Tailoring Beta-Blocker Therapy for Managing Cardiac Arrhythmias



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tBeta-blockers (BBs) are a pharmacologically diverse class, and assuming a "class effect" in arrhythmia management is an oversimplification. This illustration provides a framework for optimal agent selection based on the specific pathophysiology and clinical scenario. AF = atrial fibrillation; ARVC = arrhythmogenic right ventricular cardiomyopathy; HFrEF = heart failure with reduced ejection fraction; ICD = implantable cardioverter-defibrillator; IV = intravenous; LQTS = long QT syndrome; LVOTO = left ventricular outflow tract obstruction; POAF = postoperative atrial fibrillation; QD = once daily; SVT = supraventricular tachycardia; Tach = tachycardia.

Effects of Nebivolol Intervention on Outcomes and Rehospitalization in Seniors With Heart Failure) trial[95] found no benefit in elderly patients with HF and AF. Its unique properties (beta$_1$-selectivity and NO-mediated vasodilation) suggest potential for specific indications (eg, a role in AF prevention by modulating endothelial function), and this gap should stimulate future comparative research (Figure 2).

CLINICAL IMPLICATIONS AND FUTURE DIRECTIONS

Clinicians should prefer nonselective agents (eg, nadolol, propranolol) for patients with CPVT, LQTS, and VAs presenting with electrical storm and carvedilol for preventing POAF and for reducing atrial arrhythmias/inappropriate ICD shocks in patients with HF. The mortality class effect of BBs in HFrEF has to be considered when balancing tolerability, but for arrhythmia-specific endpoints, agent selection also matters (Central Illustration). Although challenging to implement, a large-scale RCT comparing guideline-recommended target doses of carvedilol, metoprolol succinate, and bisoprolol, with a primary endpoint of appropriate ICD therapies, might provide evidence to inform clinical practice in the management of VAs. Clinical research should also expand to explore the role of specific BBs in tailored arrhythmic indications, leveraging their unique pharmacologic properties, including the evaluation of "dual beta-blockade" strategies such as combining sotalol with HFrEF-approved BBs in patients with HFrEF, ICDs, and refractory arrhythmias.

The development of novel BBs has largely stagnated. Recent approvals, such as those for nebivolol and landiolol, involve molecules developed decades ago. Future preclinical research must prioritize "next-generation" agents, such as specific biased ligands for cardioprotection,[96,97] to maximize efficacy while minimizing extra-cardiac side effects.

CONCLUSIONS

The antiarrhythmic effects of BBs are complex and extend beyond traditional receptor selectivity. Choosing a BB should be tailored to the individual patient, the particular arrhythmia, the underlying pathophysiology, and the unique pharmacologic profile of each drug. More RCTs are needed to address current uncertainties, especially in the management of VAs.

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